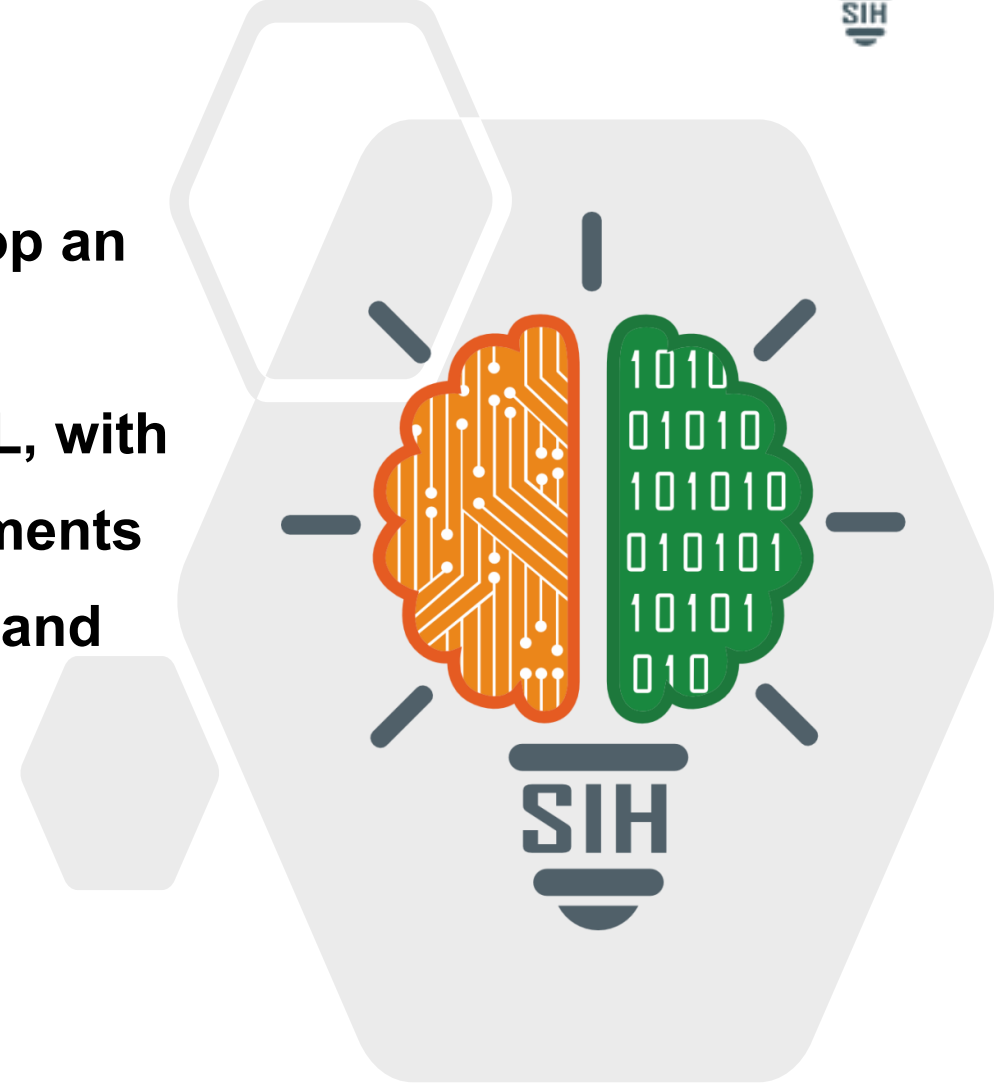


SMART INDIA HACKATHON 2025



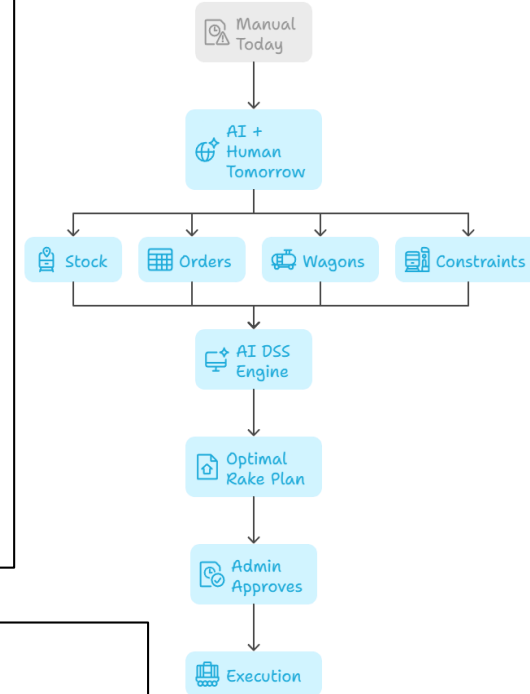
- **Problem Statement ID – 25208**
- **Problem Statement Title - Design and develop an AI/ML-based Decision Support System for optimizing rake formation strategies for SAIL, with the initial implementation focused on movements from Bokaro Steel Plant to CMO stockyards and customer location**
- **PS Category- Software**
- **Team ID - 64085**
- **Team Name : Team OG1**



OptiRake DSS



Transition from Manual to AI-Enhanced
Railway Operations



Problem Overview

- SAIL moves bulk steel, coal, and raw materials daily via rakes (full trainloads).
- Rake formation is currently manual, depending on stockyard staff, order clerks, and railway coordination.
- Challenges today:
 - Delayed rake formation → missed deadlines.
 - Underutilized wagons → partial loads, wasted freight.
 - High demurrage (penalty for idle wagons).
 - No systematic order-priority vs rake-availability balance.
- Urgent need for an AI/ML-based decision support system.

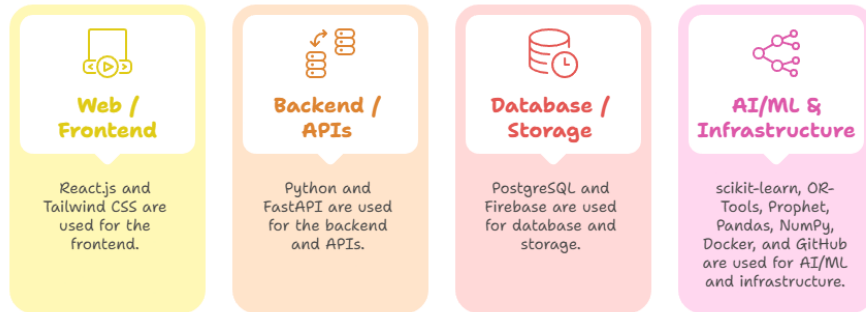
Detailed Solution

- AI/ML-based Decision Support System (DSS) for daily rake formation.
- Inputs: stock levels, customer orders, wagon availability, loading constraints.
- Optimization engine (ILP + heuristics) generates cost-optimal rake plans.
- Outputs: best rake formation (stockyard → orders → wagons → route), cost & utilization metrics.
- Plans reviewed & approved in Admin Console → linked with Traffic MIS for execution.

Innovation & Uniqueness

- Human-in-the-Loop (HITL): Admins can approve, override, or adjust → every override saved as a training case.
- Continuous learning: System improves with feedback & outcomes over time.
- Hybrid approach: Optimization solver (hard constraints) + ML forecasts (uncertain demand/delays).
- Multi-modal readiness: Rail + road dispatch optimization in future.
- Alignment with national systems: Uses PM GatiShakti, Steel Dashboard, and Traffic MIS for integration.

Tech Stack Components



AI/ML DSS Workflow for Rake Formation Optimization



DSS Use Cases

Scenario	Description
 Customer Demand & Order Fulfillment	Prioritizes wagons and full rake loading.
 Wagon Shortage	Optimizes order combination to maximize utilization.
 Multi-Stockyard Sourcing	Suggests partial loading from multiple stockyards.
 Festival/Seasonal Demand Spike	Reserves rakes and prioritizes high-margin products.
 Loading Point Breakdown	Reallocates rakes or reschedules dispatches.
 High Priority Government Order	Overrides normal optimization for first dispatch.
 Road + Rail Hybrid Fulfillment	Suggests road dispatch for partial orders.
 Delay Risk Prediction	Adjusts formation schedule to reduce idle time.
 Hazardous Material Handling	Prevents mixing hazardous goods with finished steel.
 Dynamic SLA Monitoring	Real-time alerts for dispatch at risk of missing SLA.

Feasability of idea

- Proven optimization (ILP via OR-Tools) and ML forecasting models already used in logistics.
- Compatible with SAIL ERP/SAP; works with simple CSV/Excel feeds if APIs are missing.
- Human-in-the-loop ensures admins remain in control of final plans.
- Minimal disruption: outputs integrate into existing workflows (CSV/Excel, memos).
- Economic benefit: higher rake utilization, reduced demurrage, better SLA compliance.

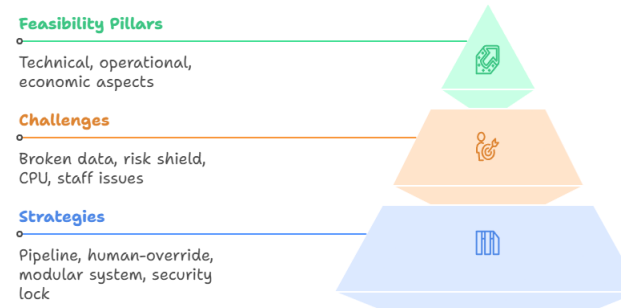
Challenges & Risks

- Incomplete or messy data from ERP and wagon logs.
- Limited real-time integration with Indian Railways systems.
- Initial ML models may miss rare edge cases or show low accuracy.
- Resistance from staff fearing automation or over-dependence on AI.
- Technical risks: performance under large order volumes, cybersecurity threats.

Strategies

- Strong preprocessing + fallback rules for missing/inconsistent data.
- Hybrid solver: ILP for hard constraints, ML only for uncertain predictions.
- Admin approval mandatory; all overrides logged as training cases.
- Pilot-first rollout (Bokaro→CMO) with modular APIs for later expansion.
- Cloud scaling + heuristic fallbacks to handle large problem sizes.
- Security: encryption, RBAC, audit logs; staff training to build trust.

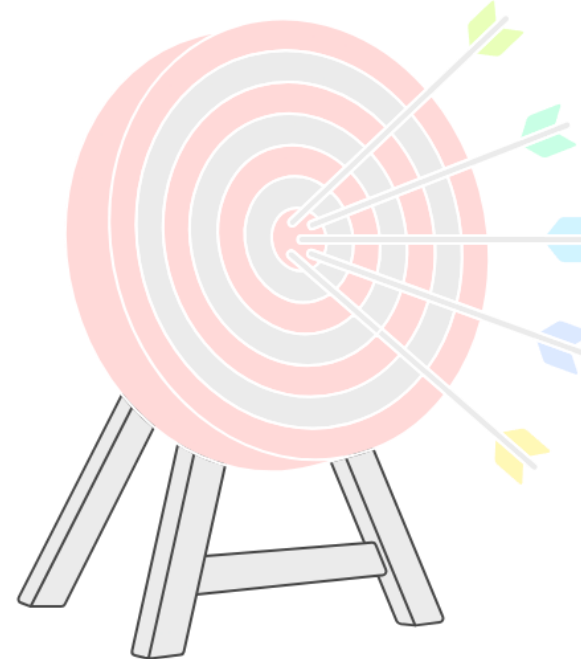
Project Feasibility Framework



Potential Impact & Benefits

- Economic: Reduced demurrage and freight cost, better rake utilization, improved ROI on wagons & sidings.
- Operational: Faster daily planning, less manual coordination, knowledge retention through logged overrides.
- Customer: On-time deliveries improve buyer satisfaction and avoid project delays.
- Social: Supports government priority orders, empowers staff with decision-assist tools.
- Environmental: Fewer partial rakes lower CO₂ emissions; better rail utilization reduces road congestion and pollution.

DSS Impact Framework



DSS Impact

Core goal of the framework



Economic Impact

Financial benefits and growth



Operational Impact

Efficiency and process improvements



Social Impact

Community and societal well-being



Environmental Impact

Sustainability and ecological balance

References & Research Work

- Ministry of Steel Annual Report 2023–24: logistics optimization is a key focus area.
- Indian Railways FOIS system: existing framework for wagon/rake availability data.
- NITI Aayog Transport Sector Report: need for multimodal (rail + road) optimization.
- IIT Kharagpur research: integer programming models for rake scheduling in Indian Railways.
- Case studies: Tata Steel and JSW Steel logistics digitization initiatives.
- Tools & methods: Google OR-Tools (optimization), Prophet & scikit-learn (forecasting).

Research & References



Top Layer

Government and policy related content. Includes MoS, Railways, and NITI.



Middle Layer

Research and academia related content. Includes IIT papers and hybrid AI-OR studies.



Bottom Layer

Industry and technology related content. Includes Tata/JSW case studies, OR-Tools, and Prophet.

Prototype Images :

